

Experimental Verification of Compact Bell-Type Inequality for 8-Qubit GHZ State on IBM QPUs

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Abstract: We experimentally investigated a compact Bell-type inequality using an 8-qubit GHZ state on IBM’s superconducting quantum processors (QPU). The reduced correlator set lowers measurement overhead, making it suitable for NISQ devices. We investigate two state-preparation strategies—a unitary gate-based chain and a star topology—on two different QPUs named `ibm_fez` and `ibm_marrakesh`.

1. Introduction

Quantum entanglement is a key resource in quantum information science, enabling tasks such as teleportation, dense coding, and quantum cryptography [1]. Bell-type inequalities provide a fundamental test of non-classical correlations by ruling out local hidden variable models [2]. While bipartite violations, such as CHSH, are well established, extending these tests to multipartite systems becomes experimentally demanding due to increasing complexity [3]. Multipartite states, including GHZ, Dicke, cluster, and W states, are essential for quantum computing, metrology, and device-independent protocols [4]. However, standard multipartite inequalities, such as MABK, require exponentially many correlators, limiting their feasibility on NISQ hardware [5]. Compact Bell-type inequalities with fewer measurement settings offer a practical alternative while preserving sensitivity to multipartite nonlocality.

In this work, we experimentally investigated a compact four-term Bell-type inequality constructed from Pauli operators ($\sigma_x, \sigma_y, \sigma_z$) on an 8-qubit GHZ state. The basis-independent operator decomposes directly into measurable Pauli correlators, enabling efficient implementation on superconducting processors. The Experiments were performed on two different QPUs `ibm_fez` and `ibm_marrakesh`, comparing chain and star-topology state preparation circuits [6]. To suppress noise, we employ dynamical decoupling [7] and measurement mitigation methods including T-REx [8] and M3 [9]. Our results demonstrate observable multipartite Bell violations on current quantum hardware.

2. 8-qubit GHZ state generation and its compact Bell-type inequality

A 8-qubit Greenberger–Horne–Zeilinger (GHZ) state is a paradigmatic multipartite entangled state that exhibits maximal quantum correlations. Mathematically, the 8-qubit GHZ state is expressed as

$$|GHZ_8\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes 8} + |1\rangle^{\otimes 8}) \quad (1)$$

Due to its strong multipartite correlations, the GHZ state is particularly well suited for demonstrating violations of Bell-type inequalities. In this work, we generate an eight-qubit GHZ state on IBM Quantum processors using two distinct preparation methods and investigate its nonlocality through a compact Bell-type inequality.

2.1. Unitary Gate Based: Chain Topology

In the chain topology, the eight-qubit GHZ state is generated by applying a Hadamard gate to qubit i , followed by a sequence of CNOT operations between neighboring qubits ($i^{th} \rightarrow (i+1)^{th}$), thereby entangling the qubits in a linear arrangement. This exploits the device’s native connectivity and therefore does not require SWAP operations, resulting in a lower circuit depth.

2.2. Unitary Gate Based: Star Topology

In the star topology, the GHZ state is prepared by applying a Hadamard gate to the central qubit 0, which subsequently acts as the control for CNOT operations targeting all other qubits ($0 \rightarrow j$). While the star topology simplifies the generation of logical entanglement and improves correlation uniformity, hardware connectivity constraints introduce SWAP overhead, increasing circuit depth.

3. Bell-type inequality for 8-qubit GHZ state

To experimentally demonstrate multipartite nonlocality while keeping the measurement overhead minimal, we employ a compact four-term CHSH-type Bell inequality. Such inequalities can be systematically constructed using the iterative

framework introduced in [10], where multipartite Bell inequalities are derived by extending lower-party CHSH-type facets of the correlation polytope. A key advantage of this approach is that it yields tight Bell inequalities involving only four correlation functions, independent of the number of qubits, making them particularly suitable for large-scale experiments.

Following this construction, the Bell inequality for an eight-qubit system takes the generic form

$$\mathcal{J} = \langle a_1 b_1 c_1 d_1 e_1 f_1 g_1 h_1 \rangle - \langle a_2 b_2 c_1 d_2 e_1 f_1 g_1 h_1 \rangle + \langle a_2 b_2 c_2 d_1 e_2 f_2 g_2 h_2 \rangle + \langle a_1 b_1 c_2 d_2 e_2 f_2 g_2 h_2 \rangle \quad (2)$$

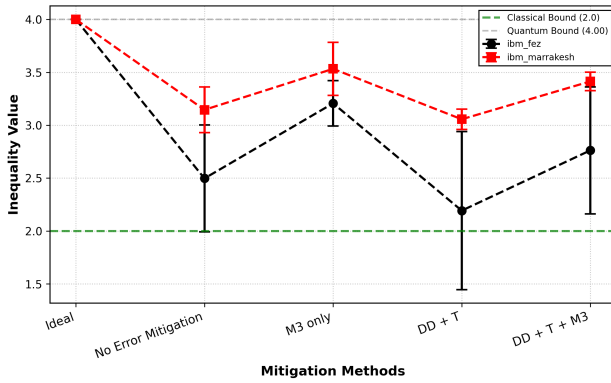
where each party chooses between two dichotomic observables. To obtain the maximal violation of this inequality for the eight-qubit GHZ state, we express each local observable as a generalized spin measurement of the form

$$x_i = \vec{n}_i \cdot \hat{\sigma} = \sin \theta_i \cos \phi_i \hat{\sigma}_x + \sin \theta_i \sin \phi_i \hat{\sigma}_y + \cos \theta_i \hat{\sigma}_z \quad (3)$$

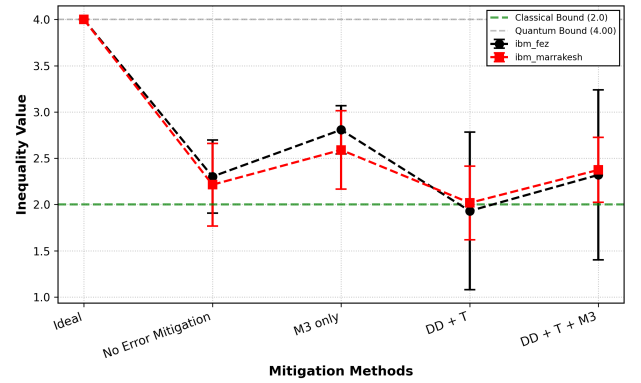
as adopted in experimental Bell tests for Dicke states [11]. Here, $\vec{n}_i$ is a unit vector on the Bloch sphere parametrized by the polar and azimuthal angles (θ_i, ϕ_i) .

4. Result and Discussion

We adopted a similar approach for extracting the results as in our previous work [12]. The quantum circuits were implemented in Qiskit, and state measurements were performed by applying single-qubit rotation gates parameterized by (θ_i, ϕ_i) prior to measurement in the computational (Z) basis. The circuits were executed on IBM Quantum hardware using the SamplerV2 Runtime Primitive. Each measurement setting was repeated three times with 10,000 shots per run to ensure reliable statistics. To assess robustness, experiments were performed on two different QPUs at different times and on different days.



(a) Expectation Value of Operators for Chain Method



(b) Expectation Value of Operators for Star Method

Fig. 1: A comparison of Bell inequality violations for an 8-qubit GHZ state prepared using two different methods on two distinct QPUs.

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